Table 4. Results of CVE vulnerability reproduction ("–reverse-limit" and "–states-limit" are two customized options CBC provides for controlling the limit of traversing the dependence graph).

Vulnerablity	Search	reverse-limit	states-limit							
			10	20	30	40	50	80	1000	5000
CVE-2012-1569	Random	4	N/A	N/A	0:44	0:57	0:52	02:03	46:20	49:01
		5	N/A	N/A	N/A	0:58	0:47	01:01	45:13	Timeout
		50	N/A	N/A	N/A	01:41	01:05	02:22	46:58	52:01
	DFS	4	N/A	0:33	0:21	01:44	0:57	02:26	02:26	02:27
		5	N/A	0:32	0:22	0:48	0:57	0:48	01:28	01:27
		50	N/A	0:32	0:24	0:49	0:59	0:56	0:56	0:57
	Coverage	4	N/A	N/A	N/A	0:26	0:36	0:28	0:53	01:07
		5	N/A	N/A	N/A	N/A	0:37	01:18	02:17	01:58
		50	N/A	N/A	N/A	N/A	N/A	01:09	01:19	06:06
CVE-2014-3467(1)	Random	4	N/A	N/A	N/A	N/A	N/A	N/A	Timeout	Timeout
		5	N/A	N/A	N/A	N/A	N/A	N/A	Timeout	Timeout
		50	N/A	N/A	N/A	N/A	N/A	N/A	Timeout	Timeout
	DFS	4	N/A	N/A	Timeout	Timeout	Timeout	Timeout	Timeout	Timeout
		5	N/A	N/A	Timeout	Timeout	Timeout	Timeout	Timeout	Timeout
		50	N/A	N/A	Timeout	Timeout	Timeout	Timeout	Timeout	Timeout
	Coverage	4	N/A	N/A	Timeout	Timeout	Timeout	Timeout	Timeout	Timeout
		5	N/A	N/A	Timeout	Timeout	Timeout	Timeout	Timeout	Timeout
		50	N/A	N/A	Timeout	Timeout	Timeout	Timeout	Timeout	Timeout
CVE-2014-3467(2)	Random	4	N/A	N/A	0:28	0:23	N/A	0:40	04:45	Timeout
		5	N/A	N/A	0:20	0:18	N/A	0:29	04:38	Timeout
		50	N/A	N/A	0:32	0:29	N/A	0:43	05:56	Timeout
	DFS	4	0:05	02:09	Timeout	Timeout	Timeout	Timeout	Timeout	Timeout
		5	0:05	02:01	Timeout	Timeout	Timeout	Timeout	Timeout	Timeout
		50	0:06	01:56	Timeout	Timeout	Timeout	Timeout	Timeout	Timeout
	Coverage	4	N/A	N/A	0:10	0:35	0:22	0:30	01:08	01:53
		5	N/A	N/A	0:15	0:17	0:19	0:23	01:29	06:13
		50	N/A	N/A	0:24	N/A	0:25	0:33	01:25	0:53
CVE-2014-3467(3)	Random	4	N/A	N/A	N/A	N/A	0:34	0:41	Timeout	Timeout
		5	N/A	N/A	0:29	N/A	0:34	0:54	Timeout	Timeout
		50	N/A	N/A	N/A	0:56	0:45	0:57	Timeout	Timeout
	DFS	4	N/A	0:06	Timeout	Timeout	0:09	0:17	0:19	0:19
		5	N/A	0:07	Timeout	Timeout	0:10	0:18	0:18	0:19
		50	N/A	0:10	Timeout	Timeout	0:12	0:20	0:21	0:22
	Coverage	4	N/A	N/A	0:35	0:41	01:53	Timeout	16:24	02:25
		5	N/A	0:13	0:43	0:30	N/A	Timeout	Timeout	Timeout
		50	N/A	0:13	0:40	N/A	N/A	0:51	Timeout	50:11
CVE-2015-2806	Random	4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	15:21
		5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	14:40
		50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	16:24
	DFS	4	N/A	N/A	N/A	N/A	N/A	20:08	Timeout	Timeout
		5	N/A	N/A	N/A	N/A	N/A	21:48	Timeout	Timeout
		50	N/A	N/A	N/A	N/A	N/A	21:55	Timeout	Timeout
	Coverage	4	N/A	N/A	N/A	N/A	N/A	N/A	02:08	Timeout
		5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Timeout
		50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Timeout
CVE-2015-3622	Random	4	N/A	0:43	N/A	0:45	02:03	02:48	05:37	04:13
		5	N/A	0:41	N/A	0:47	01:53	02:40	06:09	04:39
		50	N/A	0:55	N/A	01:03	02:31	03:27	07:29	05:28
	DFS	4	N/A	N/A	06:13	01:30	01:25	01:40	01:35	01:36
		5	N/A	N/A	06:15	01:32	01:26	01:40	01:35	01:37
		50	N/A	N/A	06:53	01:53	01:48	02:07	02:07	02:07
	Coverage	4	N/A	N/A	N/A	N/A	N/A	01:06	14:25	09:58
		5	01:14	N/A	N/A	N/A	02:12	01:19	16:14	13:21
		50	N/A	0:37	01:23	01:04	01:04	01:19	15:49	14:16

* The time format is *minute:second* and "N/A" refers to the normal termination without reproducing the vulnerability.

A Detailed Evaluation Results of CBC [59] on CVE Vulnerability Reproduction

Since the experimental results reported by the original paper (Table 2 in CBC [59]) use fine-tuned settings (the values for their customized "–reverse-limit" and "–states-limit" options) for every vulnerability in every search strategy, which we think such a running setting could be an unfair comparison. To reduce threats, we used the same various values used in the CBC paper for the two options and conducted thorough experiments on those values to investigate their impacts. We set a timeout of 1 hour and record the time for every vulnerability per search strategy.

Table 4 presents the detailed results. We can observe that there are large variations of the results under different settings. We selected the settings ("–reverse-limit=5" and "–states-limit=80") that can produce the best overall results and reported them in this paper (For the *Timeout* cases, we performed another run with an increased timeout of 24 hours for a fair comparison).